GUEST SESSION · FRESHERS' ORIENTATION

From Campus to Startup

Building products and careers for the future

Founding Engineer, Powerplay · Ex-CTO, Tornado · IIT Roorkee

Cisco → Mubble → Pramati → Tornado → Powerplay

BEFORE WE START

I had no plan. That turned out to be **the plan.**

Everything I'm about to show you was built by taking the next interesting step well; not by predicting the whole path.

01

THE ROAD SO FAR

One journey, many detours

Six stops. Not one of them was on a plan I made in first year.

THE ROAD SO FAR

Cisco → Mubble → Pramati → Tornado → Powerplay

1	2	3	4	5	6
IIT Roorkee	**Cisco**	**Mubble**	**Pramati**	**Tornado**	**Powerplay**
B.Tech	**Engineer**	**Early startup**	**Product eng.**	**CTO**	**Founding Eng.**
Fundamentals, friends, first code	How big systems are really built	First taste of 0 → 1	Products at enterprise scale	Owning tech, team & outcomes	Digitising construction

Each step just opened the next door. I only see the line looking backward.

02

UNLEARNING FIRST

Three myths to drop in your first month

Before you learn anything new, unlearn these. They quietly steer people wrong for years.

MYTH 1 OF 3

“My branch decides my career.”

Reality: your projects, skills and curiosity decide it. The degree opens the door; you choose the room.

MYTH 2 OF 3

“I need the perfect plan by year one.”

Reality: nobody has one.

Optimise for learning rate and interesting problems. Good plans are discovered by doing.

MYTH 3 OF 3

“Grades alone equal capability.”

Reality: grades keep options open but what you've built, shipped, and can explain is what teams hire for.

Do both; obsess over neither.

03

CISCO VS MUBBLE : WHAT I SAW

Big company or startup?

The question every one of you will face.
The honest answer: both teach differently!

TWO SCHOOLS OF ENGINEERING

Both are real education

BIG COMPANY (Cisco, Pramati)

- Rigour: code review, testing, processes that survive scale
- Depth in one area, world-class mentors nearby
- Stability, brand, structured growth ladders
- You own a well-defined slice of a big machine

STARTUP (Mubble, Tornado, Powerplay)

- Ownership: your code meets real users this week
- Breadth: backend, product, hiring, support; all of it
- Speed and ambiguity: you learn by deciding
- A direct line from your work to survival

There is no wrong answer; scale teaches rigour, startups teach ownership. The best engineers I know collected both.

04

PRODUCT > CODE

What “building a product” means

The most expensive lesson of my career and the cheapest to teach you now.

THE SURPRISE

Code is maybe
30% of the job.

The rest is understanding people, making trade-offs, and communicating clearly. The best engineers I know are really good at the other 70%.

THE PRODUCT LOOP

How a product actually gets built

1

Talk to users

Sit with the people who have the problem. For us: construction sites, not conference rooms.

2

Build small

The smallest version that tests the riskiest assumption. Weeks, not quarters.

3

Ship fast

Real feedback starts only in users' hands. Perfect is the enemy of shipped.

4

Measure honestly

Do people use it, return to it, pay for it? Data over opinions including yours.

5

Iterate or kill

Double down on what works. Deleting features you loved is part of the job.

This loop never ends. Product work is never 'done'; that's the job.

05

CASE STUDY · POWERPLAY

Unglamorous problems are gold mines

The biggest opportunities hide in huge, offline industries that technology skipped.

THE PROBLEM WORTH SOLVING

India builds. Most of it runs on paper.

~70 Mn

people work in construction - one of India's largest employers

~9%

of India's GDP comes from construction & real estate

Lowest

digitisation of almost any major industry; pen, paper, phone calls

Delays, material leakage and rework are the norm not because people are careless, but because the tools never arrived.

THE PRODUCT

Built for a low-end phone, a patchy network, and a real job site.

Powerplay is a mobile-first app where site teams and contractors track progress, materials, labour and money in their own language.

THE LESSON FOR YOU

Everyone chases fashionable ideas. The gold is in the 'boring' ones.

The biggest opportunities hide in huge, offline industries technology has skipped and *India is full of them!*

06

FROM TORNADO & POWERPLAY

Four lessons from the CTO seat

Leadership isn't a title you receive. It's a set of habits you can start in your first-year project team.

THE CTO SEAT

What actually moves the needle

01 People are the real system

A CTO's output is the team's output. Hiring well, giving context and unblocking others beats writing heroic code yourself.

02 Simple beats clever

Clever architecture impresses engineers; simple architecture survives 2 a.m. outages, new hires and pivots. Optimise for being understood.

03 Speed is a feature

In a startup the biggest risk is building the wrong thing slowly. Short feedback loops beat long-term perfection almost every time.

04 Stay close to users

The moment leaders stop talking to customers, the product drifts. Titles change; that habit shouldn't.

You can practise all four this year in a club, a team project, a hackathon.

07

THE ECOSYSTEM YOU'RE ENTERING

Why India, why now

You are starting college at the single best time in history to build in this country. I mean that literally.

THE SCALE OF THE MOMENT

This didn't exist when I graduated

3rd

largest startup ecosystem in the world

100+

unicorns built in India, most in the last decade

1.5 L+

DPIIT-recognised startups across 750+ districts

Figures are ballpark for 2025–26. The point stands: the market, the capital, and the rails all arrived at once.

TAILWINDS YOU GET FOR FREE

Three forces at your back

01 Digital public rails

UPI, Aadhaar, ONDC distribution and payments that took the West decades to build. You get them as a starting point.

02 A billion-user market

Real problems worth solving in every sector, in every language, at a scale that exists almost nowhere else.

03 AI as a force multiplier

Small teams now ship what needed fifty engineers five years ago. Leverage has never been this cheap.

You are starting college at the single best time in history to build in India.

07

PROOF, NOT PROMISE

India is already shipping

Not a forecast!

RECEIPTS · 2026

Five breakthroughs, five sectors

NU

Nuclear: PFBR

Indigenously built fast breeder reactor at Kalpakkam hit first criticality. **6 Apr 2026**

SP

Space: Vikram-1

Skyroot's launches India's first privately built orbital rocket. **18 Jul 2026**

PW

Power: 1200 kV

BHEL's first fully indigenous 1200 kV UHVAC transformer, type-tested. **17 Jul 2026**

IN

Infra: Chenab Bridge

The world's highest railway arch bridge, linking the Kashmir Valley. Record holder

FI

Fintech: UPI

12 bn+ payments a month. Now expanded to France, UAE, Singapore etc

Public labs, PSUs and private startups; all building at the frontier. This is the ecosystem you're joining.

08

WHAT TO INVEST IN

Skills that compound for 40 years

Frameworks expire every three years. Fundamentals appreciate. Invest accordingly.

COMPOUNDING SKILLS

The four I'd bet a career on

01 Deep fundamentals

Data structures, systems, math, first-principles thinking. Frameworks expire; fundamentals appreciate — and they're your only durable edge over AI tools.

02 Writing & speaking clearly

The highest-ROI 'soft' skill. Careers are made in design docs, emails and demos. If you can explain it simply, you understand it.

03 Learning how to learn

Every job I've had needed skills that didn't exist when I was in college. Speed of picking up new domains beats any single skill you graduate with.

04 Shipping end-to-end

Finishing is a skill. One small, ugly, working project teaches and signals more than ten brilliant unfinished ones.

These compound. A little invested now is worth enormously more in ten years.

09

CAREERS FOR THE FUTURE

Building a career in the AI era

The most common question in this room right now: does AI make my degree worthless? **Short answer: No**. Long answer, this section.

THE REFRAME

AI raises the floor, **not the ceiling.**

Boilerplate is becoming free. What gets scarcer and more valuable is judgment: knowing what to build, and whether what got built is right.

Four moves

01 Taste becomes the differentiator

When everyone can generate ten options, the person who picks the right one wins. Taste comes from studying great products and shipping bad ones.

02 Use AI as a senior colleague

Learn with it, review its work, question it don't copy-paste blindly. Master AI-assisted building and you'll out-ship whole teams.

03 Own problems, not tasks

Tasks get automated; problems don't. Be the person accountable for an outcome that role only grows in an AI world.

04 Judgment over syntax

Syntax is cheap now. The scarce, career-proof skill is deciding what's worth doing and knowing when it's done right.

AI won't take your job. A person who builds with AI and understands what it can't do very well might.

10

MAKING THESE YEARS COUNT

A simple two-year playbook

A direction, not a syllabus. The only real rule: build something every single year.

Explore → Build → Go deep → Choose ownership

1

Year 1

Explore & absorb

Nail fundamentals. Try everything clubs, sports, hackathons. Learn to code well whatever be your branch or find solutions to problems. Find seniors doing interesting things.

2

Year 1

Build & belong

Ship 2–3 real projects, however small. Join a team and contribute. Start writing about what you learn.

3

Year 2

Go deep & go real

Intern at a startup or research lab. Pick a domain and get seriously good. Build one thing strangers use or solves people's problem.

4

Year 2

Choose ownership

Convert learning into a role you want. Optimise for team & problem, not just pay. Keep one ambitious side project alive.

Treat this as a direction, not a syllabus the only rule is: ***build something every year!***

11

IF YOU REMEMBER ONLY THREE THINGS

Three ideas to carry out of this room

Forget the rest. Keep these.

01

Bias for building.

Ideas are cheap, courses are abundant, talk is free. The rarest habit in college and in careers is finishing and shipping real things.

02

Compound
people & skills.

Your network and your fundamentals grow like interest. Invest early in friends who build, mentors who care, and skills that don't expire.

03

Choose problems over prestige.

Chase problems that matter to real people sites, farms, clinics, classrooms. Prestige follows impact; it rarely works the other way.

“

The future belongs to those who build it and you have two whole years to practise!

THANK YOU

Questions?
Let's talk.

Ask me anything - startups, careers, tech, failures, IIT life, or what I'd do differently in your seat.

Founding Engineer, Powerplay · IIT Roorkee · Happy to connect